

# Félix LAPLANTE

## PERSONAL INFORMATION

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GITHUB: <https://github.com/felixlaplanteo>  
NATIONALITIES: French & Canadian

## EDUCATION

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| 2024–2026 | <b>Master in Mathematics and AI, Orsay</b><br>Joint coursework with the Probability and Statistics program, co-organized with Institut Polytechnique de Paris. | Ranked 1 <sup>st</sup> , 19.28/20 in M1, 18.28 in M2 |
| 2021–2024 | <b>Bachelor in Mathematics and Applications, Évry</b>                                                                                                          | Ranked 1 <sup>st</sup> , 19.425/20 in L3             |

## RESEARCH EXPERIENCE

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| APRIL–SEPTEMBER<br>2026 | <b>Research internship in Causality under Christophe Ambroise and Pierre Humbert, LaMME</b><br>Study of identifiability in structural causal models, with a focus on non-Gaussian settings and causal discovery. Analysis of quantum causal inflation constraints and their implications for classical causal representations.                                                                                                  |
| APRIL–SEPTEMBER<br>2025 | <b>Research internship on Joint Models under Estelle Kuhn, Christophe Ambroise, INRAE</b><br>Development of estimation and prediction methods for joint models and extension to multi-state models. Derivation of uniform high-probability bounds for survival functions, implementation of the <code>jmstate</code> package, simulation studies and analysis on real-world data, and comparison with state-of-the-art methods. |
| APRIL–JUNE 2024         | <b>Research internship in Clustering under Christophe Ambroise, LaMME</b><br>Design and implementation of a hierarchical topological clustering algorithm inspired by SVMs, <i>Spectral Bridges</i> . Partitioning into Voronoi regions, followed by grouping and automatic detection of boundaries and low-density regions using spectral methods on a local scale-invariant affinity matrix.                                  |

## AWARDS / GRANTS

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| JUNE 2024 | <b>Sophie Germain Excellence Scholarship</b><br>Also selected for the the non-cumulative <i>MixtAI DATAIA</i> Excellence Scholarship. |
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## PUBLISHED PAPERS / PREPRINTS

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| MAY 2026      | <b>A Post-Processing Conformal Prediction Approach for Conditional Coverage via Pivotal Scores</b><br><i>Félix Laplante</i><br>arXiv preprint <a href="https://arxiv.org/abs/2605.25852">https://arxiv.org/abs/2605.25852</a> .                        |
| OCTOBER 2025  | <b>A General Framework for Joint Multi-State Models</b><br><i>Félix Laplante, Christophe Ambroise</i><br>arXiv preprint <a href="https://arxiv.org/pdf/2510.07128">https://arxiv.org/pdf/2510.07128</a> submitted to <i>Statistics and Computing</i> . |
| DECEMBER 2024 | <b>Scalable Spectral Clustering Based on Vector Quantization</b><br><i>Félix Laplante, Christophe Ambroise</i><br><i>Computo</i> journal paper <a href="https://doi.org/10.57750/1gr8-bk61">https://doi.org/10.57750/1gr8-bk61</a> .                   |

## TALKS / POSTERS

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| JUNE 2026 | <b>A Post-Processing Conformal Prediction Approach for Conditional Coverage via Pivotal Scores</b> |
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| JUNE 2026      | Poster presentation at Ukrainian Mathematics Days (IHP) discussing the PIT-CP framework for conditional validity based on pivotal scores.                              |
|                | <b>A General Framework for Joint Multi-State Models</b>                                                                                                                |
| SEPTEMBER 2025 | Oral presentation at <i>Journées de la Statistique</i> (Clermont-Ferrand) introducing a unified framework for longitudinal biomarkers and multi-state event processes. |
|                | <b>Uniform High-Probability Bands for Survival Functions</b>                                                                                                           |
| JUNE 2025      | Poster presentation at <i>StatMathAppli</i> (Fréjus) deriving asymptotic uniform confidence bands under a joint modeling framework.                                    |
|                | <b>Selecting the Number of Clusters in a Clustering Algorithm</b>                                                                                                      |
|                | Oral presentation at <i>Journées de la Statistique</i> (Marseille) introducing a connectivity-based score for scalable spectral clustering.                            |

## OTHER ATTENDED EVENTS

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| MARCH 2026   | Workshop on Uncertainty Quantification and Causality (CAUSALI-T-AI, Jussieu) |
| OCTOBER 2025 | Winter School on Causality and Explainable AI (SCAI, Jussieu)                |
| JULY 2025    | Working group on Topological Data Analysis (LaMME)                           |
| JUNE 2025    | Working group on Causality (MIA Paris-Saclay)                                |

## SERVICES

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| MAY 2026 | Co-organizer of a Causality Working Group at LaMME with Pierre Humbert |
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## PACKAGES

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| PYTHON | <p><b>jmstate</b><br/>Companion package to the <i>Multi-State Models</i> paper, built on PyTorch.</p> <p><b>pitcp</b><br/>Companion package to the <i>Pivotal Scores Conformal Prediction</i> paper, built on zuko.</p> <p><b>sbcluster</b><br/>Companion package to the <i>Scalable Spectral Clustering</i> paper.</p> <p><b>fastkmeanspp</b><br/>C++-backed implementation of K-means++, with massive speedups over scikit-learn.</p> <p><b>mimisbm</b><br/>Vectorized port of the corresponding R project, implementing a mixture of multilayer stochastic block models.</p> <p><b>uniformbands</b><br/>Asymptotic uniform confidence bands for survival functions.</p> |
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## LANGUAGES

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| FRENCH  | Native |
| ENGLISH | Native |

## REFEREES

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| FORMER TEACHERS | <p>Gilles Blanchard</p> <p>Christophe Giraud</p> <p>Christine Keribin</p> |
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## INTERESTS & ACTIVITIES

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| HOBBIES | Photography, cycling, piano, hiking |
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